

Transformer Architecture

The **Transformer** is a neural network architecture introduced in the 2017 paper *Attention Is All You Need*. Unlike earlier sequence models such as RNNs and LSTMs, Transformers do not process tokens strictly one at a time. Instead, they use **self-attention** to determine how strongly each token in a sequence should relate to the others.

The main components of a Transformer are:

1. **Input Embeddings** – Convert words or tokens into numerical vectors.
2. **Positional Encoding** – Provides information about the position of each token because self-attention itself does not inherently understand sequence order.
3. **Self-Attention** – Allows each token to consider other tokens when building its representation.
4. **Multi-Head Attention** – Runs multiple attention mechanisms in parallel so the model can learn different types of relationships.
5. **Feed-Forward Network** – Applies additional transformations to each token representation.
6. **Residual Connections and Layer Normalization** – Help stabilize and improve the training of deep networks.
7. **Encoder and Decoder** – The original Transformer contains an encoder stack and decoder stack. Modern models often use only one part; for example, BERT is encoder-based while GPT models are decoder-based.

The key idea behind self-attention is the use of **Query (Q), Key (K), and Value (V)** vectors. Attention is commonly calculated as:

$$\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k})V$$

This allows the model to assign different weights to different tokens based on their relevance.

Transformers became highly influential because they can process many tokens in parallel during training and can model long-range relationships effectively. They form the foundation of many modern language models and have also been adapted for computer vision, speech processing, and multimodal AI.